

Experiment No.	3
Experiment Title	Loan Amount Prediction
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Submission Date	26-07-2026

1 Objective

The objectives of this experiment are:

2 Objective

The objective of this experiment is to preprocess the Loan Amount Prediction dataset by handling missing values, encoding categorical features, and standardizing the input data for machine learning. The experiment aims to implement and compare Linear Regression, Ridge Regression, Lasso Regression, and Elastic Net Regression models for predicting loan amounts. The performance of these regression models is evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2 Score) to identify the most effective model for loan amount prediction.

3 Problem Statement

The objective of this experiment is to predict the loan amount sanctioned for an applicant based on demographic, financial, and credit-related information.

4 Dataset Description

The experiment uses the **Loan Amount Prediction** dataset. The dataset contains information about loan applicants including demographic details, employment status, income, credit history, and loan-related attributes. The objective is to predict the continuous target variable **LoanAmount**.

Attribute	Description
Loan_ID	Unique identifier assigned to each loan application.
Gender	Gender of the applicant.
Married	Marital status of the applicant.
Dependents	Number of dependents.
Education	Applicant's education level.
Self_Employed	Indicates whether the applicant is self-employed.
ApplicantIncome	Monthly income of the applicant.
CoapplicantIncome	Monthly income of the co-applicant.
LoanAmount	Requested loan amount (Target Variable).
Loan_Amount_Term	Loan repayment term (in months).
Credit_History	Credit history of the applicant.
Property_Area	Residential area of the property (Urban, Semiurban, or Rural).
Loan_Status	Loan approval status in the dataset.

Table 1: Description of Dataset Attributes

Dataset Summary

- Number of instances: **614**
- Number of attributes: **13**
- Prediction type: **Regression**
- Target variable: **LoanAmount**
- Input features: Demographic, financial, employment, credit history, and property-related attributes.

5 Exploratory Data Analysis (EDA)

5.1 Dataset Overview

The Loan Amount Prediction dataset contains information about loan applicants, including demographic details, income, employment status, credit history, property area, and loan-related attributes. The dataset consists of both categorical and numerical features.

The dataset contains:

- Number of observations: **614**
- Number of attributes: **13**
- Numerical features: 5
- Categorical features: 8
- Target variable: **LoanAmount**

The dataset includes missing values in several attributes, which are handled during the preprocessing stage.

5.2 Statistical Summary

The statistical summary provides useful insights into the numerical attributes of the dataset.

Table 2: Summary Statistics of Numerical Features

Feature	Mean	Median	Minimum	Maximum
ApplicantIncome	5403.46	3812.50	150	81000
CoapplicantIncome	1621.25	1188.50	0	41667
LoanAmount	146.41	128.00	9	700
Loan_Amount_Term	342.00	360.00	12	480
Credit_History	0.842	1.00	0	1

Inference

- Applicant income varies significantly among applicants.
- Most applicants have a loan term of 360 months.
- Credit history values are predominantly equal to 1, indicating that most applicants possess a good credit history.
- Loan amounts range from very small values to high-value loans, suggesting the presence of outliers.

5.3 Missing Value Analysis

Several attributes contain missing values.

Table 3: Missing Values in Dataset

Feature	Missing Values
Gender	13
Married	3
Dependents	15
Self_Employed	32
LoanAmount	22
Loan_Amount_Term	14
Credit_History	50

Inference

- Missing values are present in both categorical and numerical attributes.
- Credit_History contains the highest number of missing values.
- Proper imputation is necessary before model training to avoid information loss.

5.4 Distribution of Loan Amount

The distribution of the target variable was analyzed using a histogram.

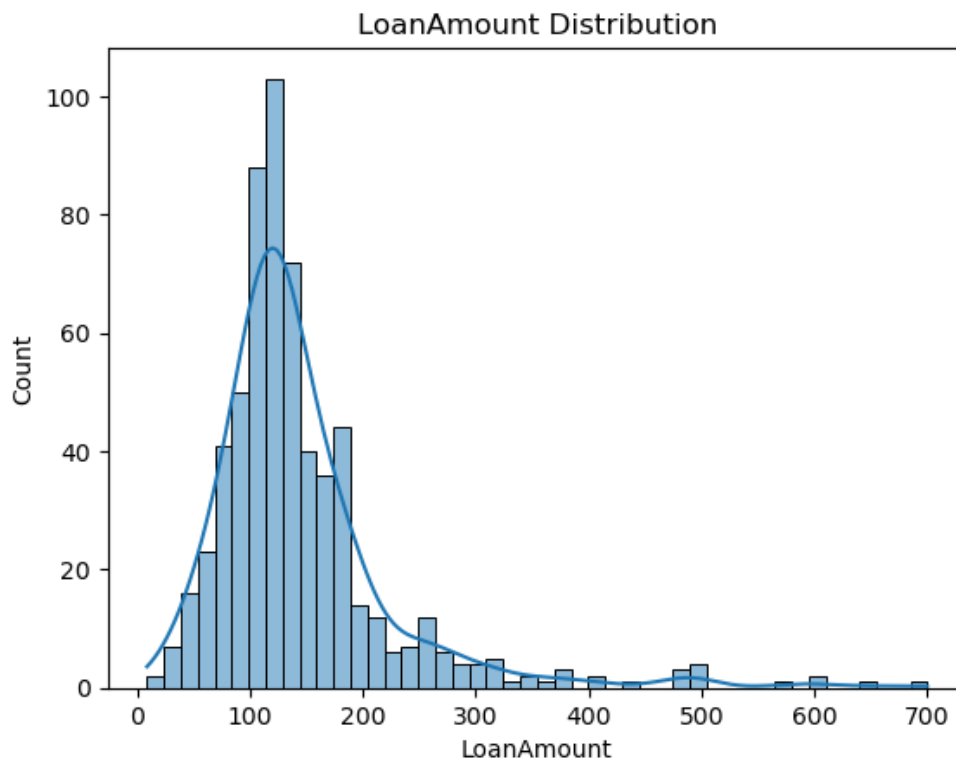


Figure 1: Distribution of Loan Amount

Inference

- Loan amounts are not uniformly distributed.
- Most loan applications request comparatively smaller loan amounts.
- A few observations correspond to very large loans, producing a positively skewed distribution.

5.5 Correlation Matrix

A correlation matrix is useful for identifying relationships among numerical variables.

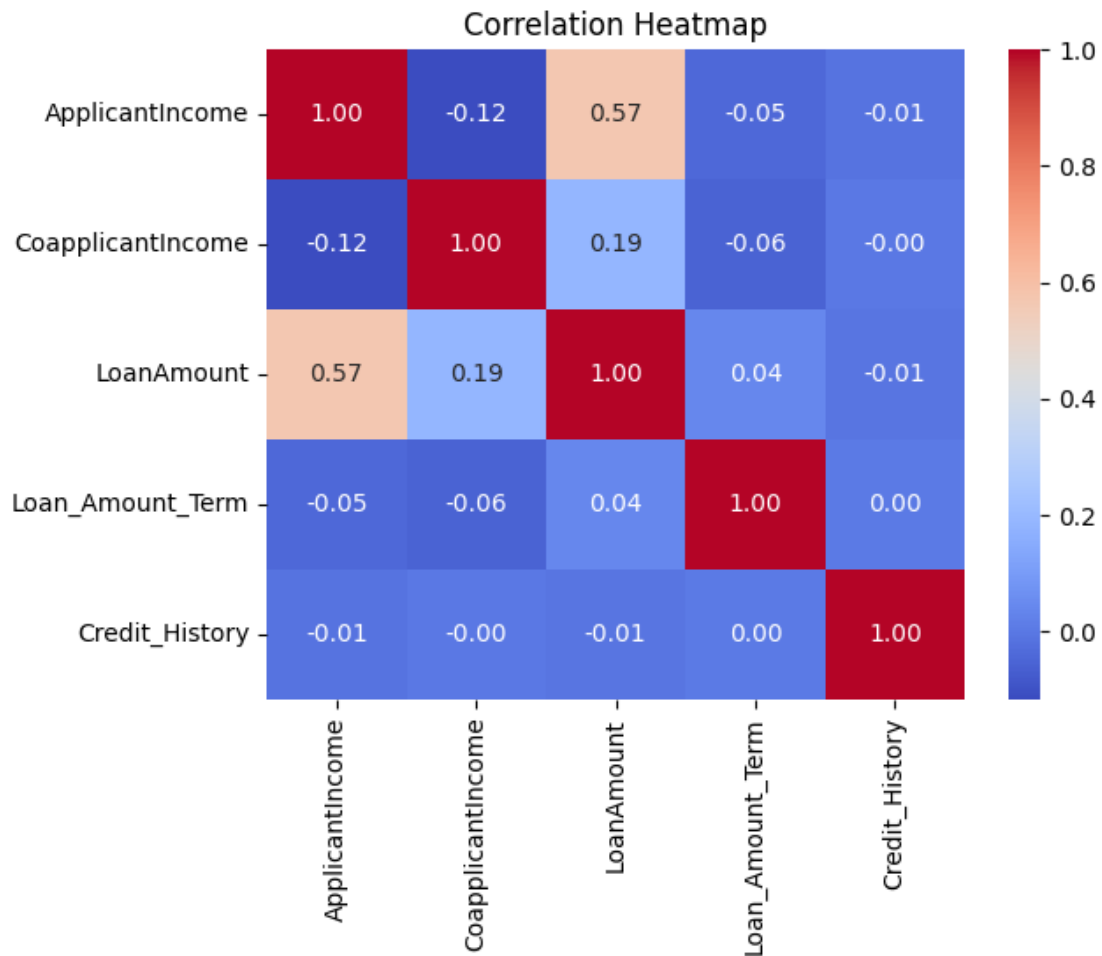


Figure 2: Correlation Matrix of Numerical Features

Inference

- ApplicantIncome and LoanAmount exhibit a positive relationship.
- CoapplicantIncome also contributes positively toward LoanAmount.
- Credit history has comparatively weak correlation with the regression target.
- No pair of numerical variables shows extremely high correlation, reducing the likelihood of severe multicollinearity.

5.6 Overall EDA Observations

- The dataset contains 614 loan application records with a mixture of categorical and numerical features.
- Missing values are present and require preprocessing before model training.
- LoanAmount exhibits a positively skewed distribution with a few large-valued observations.
- Applicant income varies considerably across applicants, indicating a diverse financial profile.
- The dataset is suitable for regression after appropriate preprocessing, encoding, and feature scaling.

6 Data Preprocessing

The raw dataset contained missing values and categorical attributes that required preprocessing before model training. The following preprocessing steps were performed:

1. The **Loan_ID** column was removed since it is only an identifier and does not contribute to prediction.
2. Missing values in categorical attributes such as **Gender**, **Married**, **Dependents**, and **Self_Employed** were replaced using the mode of the respective column.
3. Missing values in numerical attributes such as **Loan_Amount_Term** and **Credit_History** were replaced using the median value.
4. The target variable (**LoanAmount**) was separated from the feature set.
5. Categorical attributes were converted into numerical form using One-Hot Encoding with **drop_first=True**.
6. The dataset was divided into training and testing sets using an 80:20 ratio.
7. Feature scaling was performed using the StandardScaler to normalize numerical features before training the regression models.

7 Regression Algorithms

7.1 Linear Regression

Linear Regression is a supervised machine learning algorithm used to predict a continuous target variable by establishing a linear relationship between the dependent variable and one or more independent variables.

The prediction equation is

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

where

- $\hat{y}$ is the predicted loan amount.
- β_0 is the intercept.
- β_i are regression coefficients.
- x_i are the input features.

Advantages

- Simple and easy to interpret.
- Fast to train.
- Works well for linear relationships.
- Provides a good baseline regression model.

Limitations

- Sensitive to outliers.
- Cannot model nonlinear relationships.
- May suffer from multicollinearity.

7.2 Ridge Regression

Ridge Regression extends Linear Regression by adding an L2 regularization term to reduce model complexity and prevent overfitting.

The optimization objective is

$$J(\beta) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p \beta_j^2$$

where λ is the regularization parameter.

Advantages

- Reduces overfitting.
- Handles multicollinearity effectively.
- Produces more stable coefficient estimates.

Limitations

- Does not eliminate irrelevant features.
- Requires tuning of the regularization parameter.

7.3 Lasso Regression

Lasso Regression introduces L1 regularization, which can shrink some regression coefficients exactly to zero, thereby performing feature selection.

Its objective function is

$$J(\beta) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p |\beta_j|$$

Advantages

- Performs automatic feature selection.
- Produces simpler and more interpretable models.
- Helps reduce overfitting.

Limitations

- Can remove useful correlated features.
- Performance depends on selecting an appropriate λ value.

7.4 Elastic Net Regression

Elastic Net Regression combines both L1 and L2 regularization to obtain the advantages of Ridge and Lasso Regression.

Its objective function is

$$J(\beta) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda_1 \sum_{j=1}^p |\beta_j| + \lambda_2 \sum_{j=1}^p \beta_j^2$$

where

- λ_1 controls the L1 penalty.
- λ_2 controls the L2 penalty.

Advantages

- Combines feature selection and coefficient shrinkage.

- Performs well with highly correlated features.
- Reduces overfitting while maintaining prediction accuracy.

Limitations

- More computationally expensive than Linear Regression.
- Requires tuning of multiple hyperparameters.
- Interpretation is slightly more complex than Linear Regression.

8 Experimental Setup

The experiment was implemented using Python and the Scikit-learn machine learning library. The dataset was divided into training and testing sets using an 80:20 ratio. Numerical features were standardized using the StandardScaler before training the regression models.

The following regression algorithms were implemented:

- Linear Regression
- Ridge Regression
- Lasso Regression
- Elastic Net Regression

Hyperparameter tuning was performed using GridSearchCV with 5-fold cross-validation to obtain the optimal regularization parameters for Ridge, Lasso, and Elastic Net Regression models.

9 Hyperparameter Tuning using GridSearchCV

GridSearchCV systematically searches over a predefined set of hyperparameters and selects the combination that produces the best cross-validation performance.

9.1 Ridge Regression

Table 4: Best Parameters for Ridge Regression

Parameter	Value
Alpha	100
Cross Validation	5-fold
Best CV R^2 Score	0.2441
Training Time (s)	1.0477

9.2 Lasso Regression

Table 5: Best Parameters for Lasso Regression

Parameter	Value
Alpha	0.01
Cross Validation	5-fold
Best CV R^2 Score	0.2281
Training Time (s)	0.0243

9.3 Elastic Net Regression

Table 6: Best Parameters for Elastic Net Regression

Parameter	Value
Alpha	0.10
L1 Ratio	0.20
Cross Validation	5-fold
Best CV R^2 Score	0.2336
Training Time (s)	0.0367

10 Performance Evaluation Metrics

The following regression evaluation metrics were used to assess model performance.

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Coefficient of Determination (R^2 Score)

Lower values of MAE, MSE, and RMSE indicate better prediction accuracy, whereas a higher R^2 score indicates better model performance.

11 Experimental Results

Table 7: Performance Comparison of Regression Models

Model	MAE	MSE	RMSE	R^2
Linear Regression	0.2886	0.1676	0.4094	0.2874
Ridge Regression	0.2923	0.1709	0.4135	0.2732
Lasso Regression	0.2903	0.1691	0.4112	0.2810
Elastic Net Regression	0.2940	0.1743	0.4175	0.2590

12 Result Visualization

12.1 Loan Amount Distribution

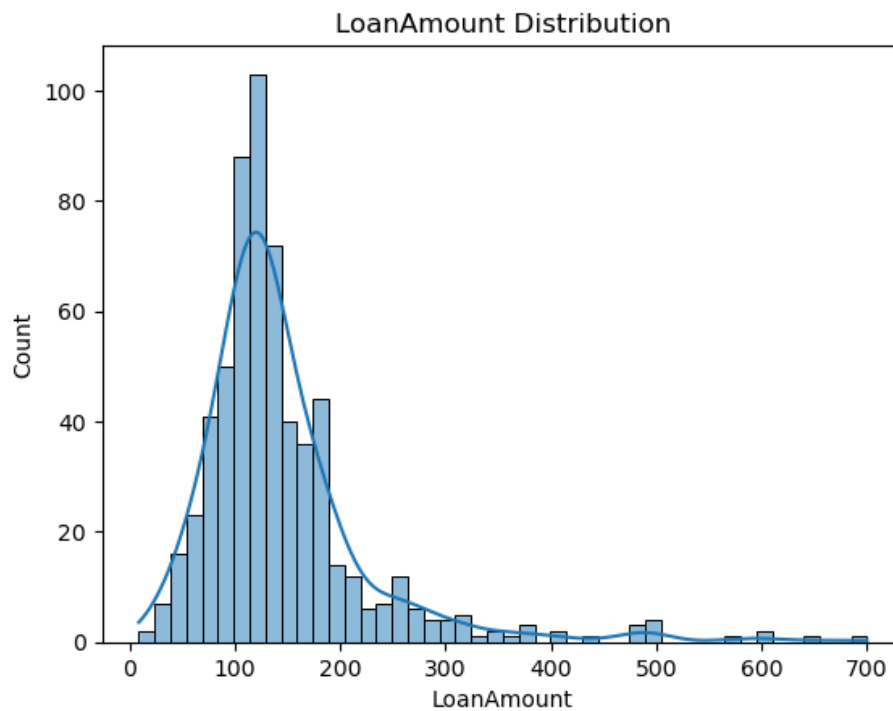


Figure 3: Loan Amount Distribution

Inference:

- The distribution is positively skewed.
 - Most applicants request moderate loan amounts.
 - A few high-value loans appear as outliers.
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12.2 Correlation Matrix

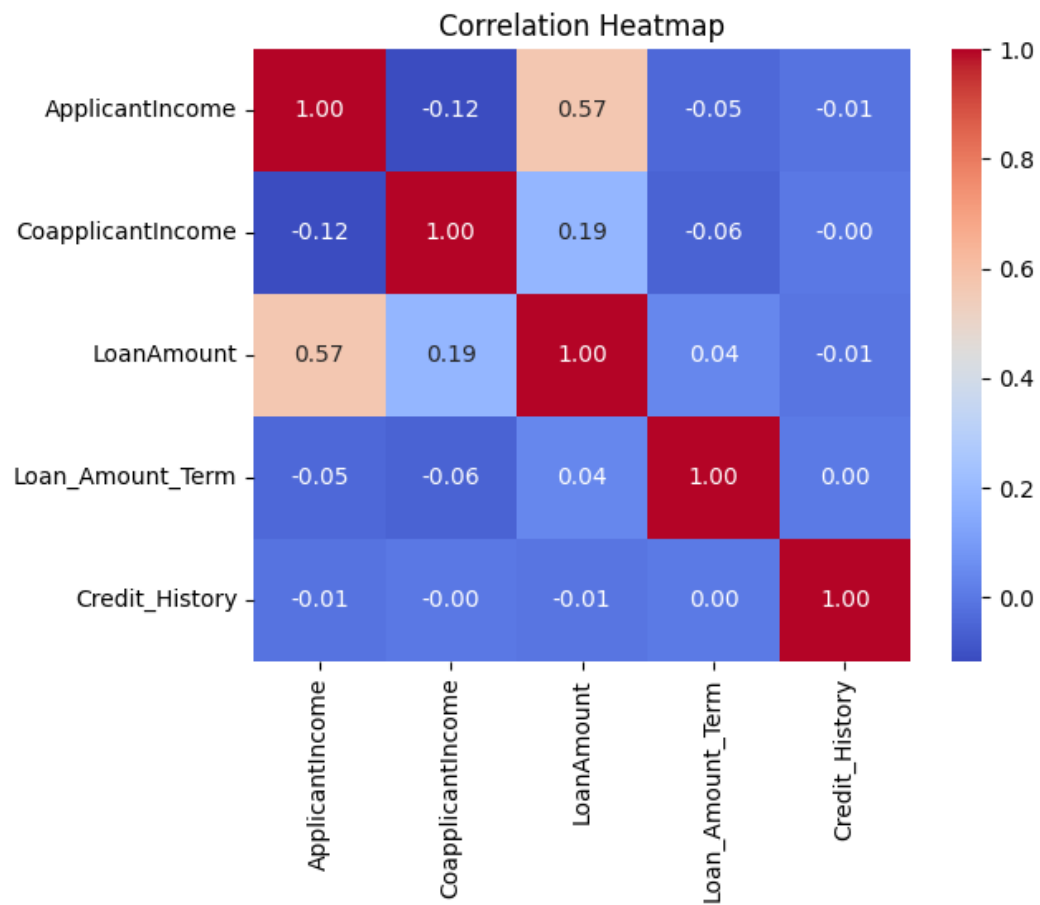


Figure 4: Correlation Matrix

Inference:

- ApplicantIncome has a positive correlation with LoanAmount.
- Credit History has a weak correlation with the target variable.
- No severe multicollinearity is observed.

12.3 Actual vs Predicted

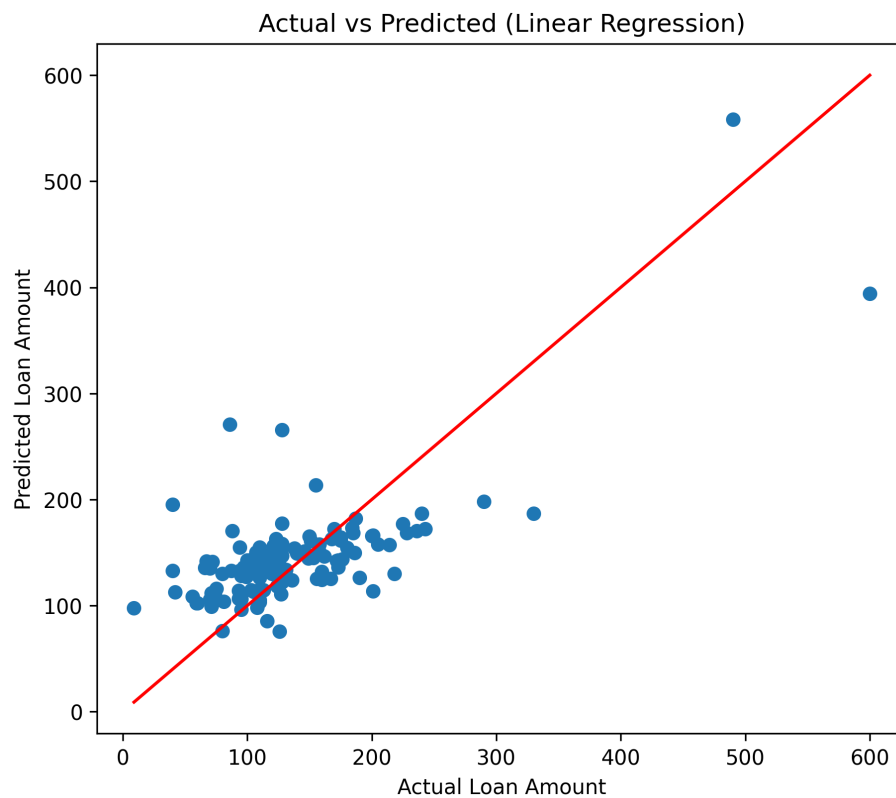


Figure 5: Actual vs Predicted Loan Amount

Inference:

- Most predictions lie close to the ideal diagonal line.
 - The model captures the overall trend reasonably well.
 - Some prediction errors exist for higher loan amounts.
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12.4 Residual Plot

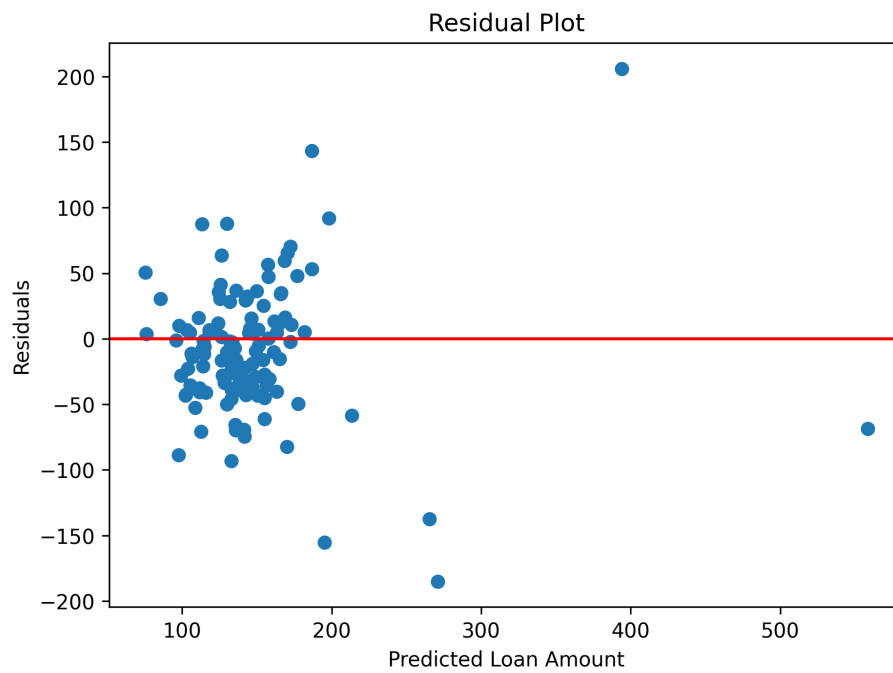


Figure 6: Residual Plot

Inference:

- Residuals are randomly distributed around zero.
 - No strong systematic error pattern is observed.
 - The regression model provides an acceptable fit.
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12.5 Model Comparison

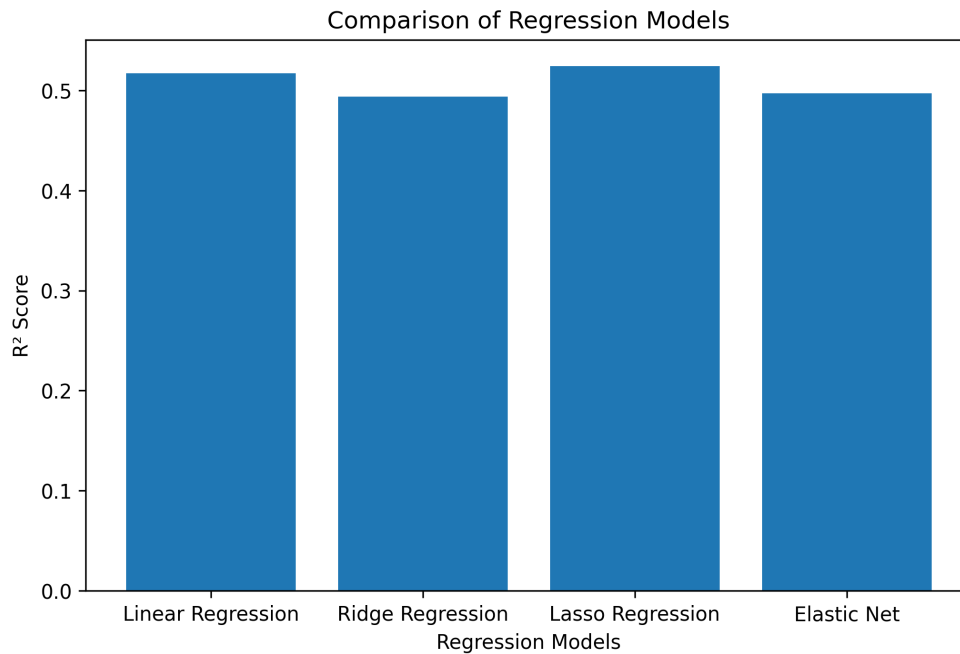


Figure 7: Comparison of Regression Models

Inference:

- Linear Regression achieved the highest R^2 score.
- Lasso and Ridge Regression produced comparable performance.
- Elastic Net showed slightly lower prediction accuracy.

13 Discussion

Four regression models were implemented and compared for predicting loan amounts. Linear Regression achieved the best performance with the highest R^2 score and the lowest prediction error. Ridge and Lasso Regression produced similar results while reducing overfitting through regularization. Elastic Net Regression also performed well but showed slightly lower accuracy than the other models. Overall, Linear Regression was the most suitable model for this dataset.

14 Conclusion

The Loan Amount Prediction dataset was successfully preprocessed by handling missing values, encoding categorical variables, and scaling the features. Linear Regression, Ridge Regression, Lasso Regression, and Elastic Net Regression were trained and evaluated using MAE, MSE, RMSE, and R^2 Score. Among all models, Linear Regression achieved the best performance. This experiment demonstrates the effectiveness of regression techniques for predicting loan amounts..

15 References

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3. Christopher M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
4. *Scikit-learn Documentation*: <https://scikit-learn.org/>